Three Ways to Destroy a Sound

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ABSTRACT

This paper presents a set of three different designed objects which each take a different approach to the same concept - the active destruction of sound recordings. I situate this work within sustainability studies, political ecologies of technology, and constraints-based design. Following this I describe the design and development of the devices themselves, introduce the final built objects, and attempt to analyze what they might represent. The paper takes the form of an annotated portfolio, in which the objects are shown, analyzed, and critically evaluated.

CCS CONCEPTS

• **Human-centered computing → Sound-based input / output**; HCI theory, concepts and models; *Interaction design theory, concepts and paradigms.*

KEYWORDS

sound, memory, technology, destruction, deleting, sustainability, time, alternative design

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1 INTRODUCTION

As a 41 year old who was first introduced to digital technologies as a child, I have already lived through at least four generations of lost media. Floppy disks (let us generously group the two different floppy disks together), zip drives, DAT tapes, and writable CDRs are all now, to various degrees, effectively lost media formats that are at best difficult to access and at worst degraded entirely. It is not difficult to imagine the millions of photographs, sound recordings, videos, and documents around the world that are lost despite being "backed up" onto these supposedly permanent media. We were promised Pure Perfect Sound Forever [10].

This is not an unrecognized problem - an entire community of "media archaeologists" work, such as in the videogame community [1], towards finding methods to get obsolete software and hardware running again so the original experiences can be preserved. And yet despite these efforts, and the recognition of past mistakes, we are currently living through another cycle of promise and loss. As we enter the twilight years of the spinning hard drives, awaiting the day when our Time Machine or Drobo backups will slowly degrade or suddenly fail, our files and memories are now being outsourced to a mysterious "cloud". This time, we don't need to worry, the cloud is limitless and invisible and all-powerful.

If we again take the example of videogames, this scenario does not bode well. The move towards trading control for convenience, accepting software updates and walled garden app stores in exchange for stability and security, has meant the disappearance of an untold number of games that did not meet upgrade requirements or simply pay the fee to be listed on app stores. I, personally, now have virtually no way to play any of the iOS videogames that I developed and released in the early 2010s. These too have been lost (though perhaps in that particular case it might be for the best).

It is effectively guaranteed that the cloud services that are now generally accepted as a “permanent" storage solution will one day fail us. It is difficult to guess precisely how - perhaps general system failure linked to climate change, combined with changing company policies or surprise bankruptcies or software updates or viruses [18] - the list is endless but the outcome is clear.

The illusion of data permanence sits neatly within the current paradigms of technology design, in which limitless options and storage and speed are all presented as the ideal. This so-called “cornucopian" [27] approach to technology mirrors the general glorification of economic growth [9, 16, 31], and is equally unsustainable. We simply can not depend on endless server space - while it may seem unimaginable now, resources are never limitless. On a micro scale this is already noticeable in the experience of constant warnings: “you are out of cloud storage space, upgrade your plan to keep your files". We expand to fill whatever storage space we have (helped by rapidly growing media file sizes due to higher quality capturing devices), far outpacing whatever storage we have available to us.

I have made three devices that explore these issues through the medium of digital sound. As a group I refer to these as the Sound Destruction Machines. The three devices can be summarized thus:

- **The Memory Boxes**: A set of wooden boxes which contain an audio file. Pressing a button simultaneously plays and permanently destroys the sound
- **The Kopiator**: A device which records sound and immediately plays it back and destroys it on loop until it disappears
- **The Sound Destroyer**: A interface that provides fine grained control for how a sound can be recorded, looped, and destroyed

This paper is about these devices, how they work and what they might mean. I will first provide some background on the design, sound, and sustainability concepts that have informed the projects. Following this I will introduce each of the devices, describing them and offering some critical analysis of what they do and represent.

2 BACKGROUND

2.1 Alternative Designs

The creation of the devices described here is quite clearly the result of a Research Through Design process [3, 14, 34], in which the manifestation of the concepts (in this case both the code and the hardware) is prioritized and is seen to contain knowledge in of itself. In particular I draw on Pierce’s definition of *alternative designs* [26] which “prefigure possible, though improbable productions" and resist further development. The friction that I encountered in the development of these designs, and which continues in their realized states, is precisely the point and I intend to avoid the caveats and qualifications that so often accompany this type of work. Rather this paper is perhaps best seen as a form of annotated portfolio [5, 13], in the sense that it describes a set of designs and attempts to tease out a set of themes and “family resemblances to be reasoned about". A paper that describes a set of physical, tangible objects will necessarily be somewhat limited, and is a poor subsitute for holding and listening to an instrument as it destroys a cherished memory or destructively moulds a sound in an unexpected and delightful way. This paper is thus accompanied by a video file showing all three of the instruments as they destroy sounds in their own unique ways, as well as a demonstration of the underlying code running an demonstration on a desktop computer environment. The video and this paper together function as the annotated portfolio, and at NordiCHI will be presented alongside the devices themselves for people to interact with.

2.2 Political Ecology of Sound

Modern sound technology relies on the paradigm of reproducibility - the ability to record, revisit, and endlessly reproduce sound “non-destructively". Tara Rodgers and others have analyzed how the language used for these processes is not accidental [29], but rather reflects how recording technology reflects and perpetuates hegemonic norms; the ideal sound “reproduction" medium is generally considered to embody feminine-coded traits of silence, passivity, and invisibility. Meanwhile, an audio source is often described as “punchy", “clear", or “strong" - and discussions of audio fidelity are generally positioned as the struggle for these masculine traits to be perceived despite the weaknesses of the reproductive surface. These analysis reveals a political ecology of recorded sound, which can be traced back to the use of disembodied human ears gathered from nameless cadavers to build the first sound recording devices [11], continuing through to gendered approaches to design and manufacturing[33]and recently with the application of deep-sea oil drilling technologies to auto-tuning [28, 30].

It is no wonder, then, that there is an embedded expectation for sonic technology to be endless, seamless, and neutral - the aforementioned cornucopian attitude to design and technology that is well integrated into all computing technology.

From a sound design perspective the cornucopian design mentality can be seen not only as harmful from a social justice and material sustainability standpoint but also philosophically problematic. Sound as a medium is uniquely unsuited to cornucopian design principles, due to its fundamentally time-based nature. As Chion and others have explored, sound is completely reliant on (and therefore particularly illustrative of) time [8], and as such listening back to a sound requires inhabiting the same amount of time as it took to record it. Although a library of photographs or digital videos can be perceived at a glance, a hard drive full of sound recordings could take hours to listen to, if not significantly more. Recording sound is effectively an attempt to force a wedge into the continuation of time, on the problematic assumption that we will one day be able to simply make the time to listen back. Thus if sound is inherently linked to our perception of time, why then does modern technology and interaction design insist on breaking this paradigm, relying instead on a cornucopian paradigm that robs sound of the exact thing that makes it powerful?

To work with sound is to confront the issues faced by the world as a whole - a problematic relationship between the past and the present, as well as our attempts to control an uncertain future. Sound is thus uniquely positioned as a medium of experience and expression with regards to sustainability, in the sense of physically embodying the effects of time. Listening to a sound is the act of physically experiencing the consequence of a release of energy in the past, just as the seemingly intractable issues of sustainability

stem primarily from our current experience of actions taken in the past.

Perhaps, then, we must reject the core assumption of recorded sound - that it exists in order to allow us to hear a sound again. What if we could only ever hear a recording once, or indeed if the very act of listening to a recording was the same act that destroyed it? If a recorded sound is no longer reproducible, perhaps it goes some way towards challenging the norms of listening and experiencing sound, as well as the greater notions of political ecologies, limitless growth, and endless storage that are so embedded into technology design and media consumption.

2.3 Limits

Within the Sustainable HCI (SHCI) community, cornucopian design has been heavily critiqued from a sustainability perspective [7], with a number of designers and researchers proposing a “limits" or constraints-based approach [17, 19, 21, 25, 32] to move towards a more socially and technologically sustainable future. This project takes this concept and considers how it overlaps with our use of media and storage. As with most of the issues in sustainability studies, the facts are clear (it is already too late to stay below 1.5°, and infinite data storage is impossible) even if the calls to action are confusing at best (recycle your straws and buy more cloud storage).

I would like to position the work in this paper alongside other HCI approaches to the overwhelming mass of media and the heavy weight of history that goes along with it. The three devices described here are referencing projects like the OLO Radio [23], which both encourages (and to a certain extent problematizes) a listener’s relationship with their past, and the tradition of autodestructive music and art typified by Marclay’s *Record Without a Cover* or even Lucier’s *I’m Sitting In A Room* [15, 20] among others. There are also echoes of the research projects looking into the vast array of digital content being left behind when people die [6, 22], and by extension how designers can consider time [12]. Where those projects consider the digital artifacts people want to keep forever, SHCI research points out that our digital footprints necessarily need to shrink (alongside our more recognizably material footprints).

This project thus accepts that we must move towards deletion. As designers we need to consider how we can provide the tools to delete and explore the experience of destroying media. The Sound Destruction Machines represent an approach that embraces the limits in order to focus on intentionality and meaning. They are here to help.

3 HOW TO DESTROY A SOUND

3.1 The sound destruction engine

The three devices in this set all use the same underlying system for destroying the sound. As such in some ways they represent how the presentation and interaction are what actually create the experience (more so than the technical system), but nevertheless let us describe how it works.

In order to position the work within contemporary media consumption and technology design the project focuses exclusively on digital sound files. The act of sonic destruction is accomplished through a wavetable playback system that overwrites the soundfile. This allows for the destruction to be permanent, constant, to a certain degree controllable, and arguably aesthetically pleasing (or at least understandable).

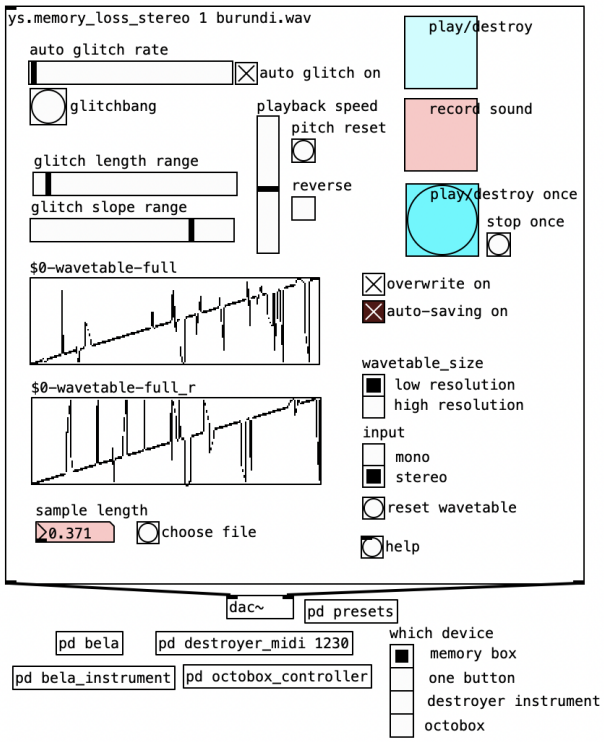


Figure 1: The sound destruction engine, destroying a sound

A sound file exists on a digital storage medium, most likely a flash disk. For the purposes of this engine it is comprised of 44100 data points per second of sound, stored in an array, each point representing a position over time of the reproduction medium that will eventually be driven to make the sound audible.

In order to drive the playback of this sound file, a second array is created that represents time. This is essentially a graph that begins as a line running from 0 to 1 over the same length of time as the source sound. Applying this graph to the audio file results in a linear playback of the sound - this is known as a wavetable.

Adjusting the wavetable to be something other than a diagonal line introduces variability into the playback of the sound file. If a section of the line has a negative slope then the corresponding section of sound will play backwards. Sharp slopes will play the sound file more quickly, and a horizontal line will stop playback entirely. All of the resulting sound is still being generated from the original source media data - introducing a number of “errors” to the wavetable can be compared to cutting up and re-assembling a photograph.

The destruction engine can introduce these errors, or “glitches", into the wavetable automatically using a set of semi-randomized parameters. Two controls are available in the code to control the size and characteristics of a glitch, dictating the range of random numbers that will be used to generate the width (time) and height (playback rate) of a glitch somewhere in the wavetable. These controls

are set and then the resulting glitches are generated and applied on command. The glitch command can be triggered individually or be scheduled to occur at a given rate (relative to the length of the original source file). The rate of playback can also be controlled, including whether it is played forward or in reverse.

In order for the destruction to be permanent, when the audio is played back the output is routed to overwrite the original file with a short delay (to allow for the playback and overwriting to happen simultaneously). In the engine this feature is switchable, so that the glitched playback can occur with or without overwriting the audio file. This method for making the destruction of the sound permanent can only occur while the sound is being played back, since it is the sonic output that is being stored into the array. The listener therefore hears the destruction as it happens.

The bulk of this paper will address what happens when this system is given form through interaction. Each of the three devices described below reveal different aspects of the destruction engine to the user through interface elements, and other aspects to obscure or control automatically. Naturally, each of these decisions changes how the resulting interaction takes shape and what type of device is generated.

3.2 The Memory Box

The first device that uses the destruction engine is The Memory Box, a device designed with the aim of generating meaningful sonic destruction. As such it requires a meaningful sound recording in order to function. Related to this, the sound contained within the Memory Box needs to be the only remaining copy of that sound recording. Only with these two rules satisfied can the device provide meaningful destruction of the sound.

I have built three of these devices. They are hexagonal wooden boxes that plug into a mains power outlet and contain a single visible button, which glows white when ready. Some programming work is required to load a sound into the electronics inside. When ready, pressing and holding the button activates playback, which is heard through a built-in speaker that vibrates the box. The sound plays for as long as the button is held, and a glitch is introduced on each loop. The overwrite function is activated in the code, so each time the sound is heard and a glitch is introduced the change is permanent. It is difficult to estimate how long a sound will last in the Memory Box, but after roughly 100 loops the original sound has been completely lost and all that remains is slowly fading noise.

What constitutes a meaningful sound? If I open my cloud photo library I can instantly listen to the sounds that accompany 1,215 video files recorded over the past ten years. Presumably I could randomly and blindly choose any one of those and delete it, and I would not ever notice it was gone. However it is equally likely that if I opened one at random to actually watch (something approaching the Photobox mechanic [24]) I would become unreasonably attached to that video and the feeling of deleting it permanently would land somewhere between disappointment and rage. The personal attachments I have to these files are, then, perhaps recreated whenever I perceive them - however this is something that I very rarely do. I simply do not have the time or mental capacity to browse 1,215 files.



Figure 2: Three Memory Boxes

How, then, would I go about selecting a sound that would be destroyed and whose destruction would carry meaning and weight? Even more challenging, how can I work with someone else to help them select a sound for themselves? And if we do manage to find a sound that they are willing to consider permanently destroying, how do they accept this?

Ensuring the uniqueness of the sound recording in the Memory Box is something of a challenge, due to the nature of modern file storage and sharing. Deleting a video file from an iCloud photo library, for example, simply moves it to a password-protected “trash" which is is only “emptied" after 30 days unless you do so manually. Even if this trash is deleted there is no guarantee that it does not continue to exist on a server, and any number of permutations exist wherein the file will not be removed from some devices (if files sync locally and are cached on a backup, for example). Messaging a file will result in multiple copies stored on various servers; cloud storage systems bias strongly against permanent deletion; the list goes on.

Due to these factors, the loading of the sound into the Memory Box devices is an action that can only be performed by me, the designer. It requires a set of relatively specialist knowledge regarding how the code needs to be set up and how the sound is loaded onto the machine. I therefore become a mediator of the media. I take a sound, prepare it for the device, load it on, and ensure that it is deleted everywhere else. Only then does it begin to function.

The outcome of these design choices is a relatively passive device that contains a sound that can only be heard if it is also simultaneously destroyed. The device can sit quietly and never be used, preserving the sound and becoming something of a visual representation of the sound file that exists only on the memory chip inside. Where photos can be printed and hung on the wall, this device offers the potential for a sonic memory to form a part of the ambient decoration of a space. However the soft glowing light and gentle form also comes as something of a threat - pressing the button, and listening to the sound that is generated, is a permanently destructive act.

Giving The Memory Box to others to use would require a careful, and very personal, approach. One scenario for giving these devices



Figure 3: The Kopiator



Figure 4: The Sound Destroyer

to others could involve working very closely with two volunteers, each of us having one of the devices. We would each explore our media libraries and identify a recording to be placed in the Box. If each person feels comfortable sharing, we could listen to each other's choices and discuss why it is meaningful. I would then "put" the recordings in the boxes and we would work together to ensure that every possible copy of the recording has been deleted elsewhere to the best of our knowledge. We would then say goodbye to the sounds, and each of us would take the devices home to live with them for some time.

3.3 The Kopiator

The second implementation of the destruction engine is designed for use in an environment with many people passing through, perhaps a hallway or lounge or similar liminal space. A prominent microphone protrudes from an otherwise nondescript metal enclosure that would not feel out of place in a conference room or lecture hall. Below the microphone is a single glowing button. The microphone will record any sound that is made while the button is held down, and the system will play the resulting recording on loop the moment the button is released. The sound will continue to play on loop, with the auto-glitch system in the software introducing errors on each playback, which are permanently written to disk. The glitch settings are high enough that the sound is destroyed and completely disappears within a couple of minutes. Pressing the button quickly at any time will stop the playback.

Where the Memory Boxes function as a representation of the past, the Kopiator is primarily a reflection of the immediate present. Any sound that is recorded represents the now, and the resulting playback is a sonic manifestation of the time period the button was held down. Any sound that occurred during that time is then played back and repeated until it disappears. Thus the device blurs time - a sound which would normally have already become a part of the past is instead prolonged. It still fades away, but it takes a bit more time and the listener is much more aware of the mechanism for how it is destroyed.

It is also, effectively, a toy. It lends itself to funny interactions, especially as people first learn to use it. With no instructions on the device, it can take a few button presses before people understand how it works (if, indeed, they figure it out at all). This often takes the form of someone saying "what does this do?" as they press the button, resulting in a fragment of the sentence being immediately played on loop with compounding errors "this do? this do? thiis do? tehisd do? ttechissdo? tetechischooo?"

3.4 The Sound Destroyer

Considering this work as a set of devices that put form to the same underlying code, it is tempting to imagine a sort of gradient between two extremes of control. Perhaps on one side is a physical manifestation of the code which is entirely automated with no user control whatsoever. Arguably this would be so extreme as to cease to function, since by removing any human interaction there would be no method for generating or selecting a sound. One step into the gradient brings us to something like the Memory Box, where a single button applies a number of predetermined actions. This creates a highly personalized experience which requires no prior training or knowledge but also offers no control (apart from the choice of whether or not to push the button) and relies on the mediation of an expert.

On the other extreme, then, we can make every control available. The resulting object begins to function as something of an instrument, in both the musical and scientific sense. The Sound Destroyer has four knobs, seven buttons, and an LED which indicates whenever a sound has been recorded or a glitch implemented. A footswitch input allows for hands-free recording and looping and audio is routed using input and output jacks.

Simply the existence of these jacks immediately moves this into a different category of interaction - they require (or assume) a level of existing infrastructure and by extension knowledge. In order for the instrument to function as a sonic object the user requires a microphone, possibly a preamp, and some kind of amplification or listening system. There is no volume control on the inputs or outputs, this has been effectively outsourced to the user along with an assumption that they have the knowledge for how an input signal can be generated and controlled and an output signal managed and routed.

The knobs and buttons are labeled and mapped to each of the available parameters in the code. While a number of these (record,

play, forward/reverse, etc) are digital signals that have only two possible states, the potentiometer knobs are analog inputs that can send a range of values. These values are then scaled to an appropriate range for the associated parameter. The total control that the instrument suggests is thus something of an illusion, since this scaling must be decided at some stage - the autoglitch speed parameter, for example, ranges from 1-10x per loop. Why not 50? 100? 1000? Why can the playback pitch knob go from 0-4?

These decisions, made by myself based on my own opinions and aesthetic desires, can be seen as analogous to tuning an instrument. A flute would vibrate a column of air in exactly the same way with a wide range of placement and number of holes, but their precise location has been chosen for aesthetic purposes, and after many generations the specific design has crystallized and standardized. The Sound Destroyer instrument has nowhere near the maturity of even a relatively new instrument, and if it were to develop further than perhaps the “tuning" of early version like this would be considered frightfully old fashioned.

This is an instrument that is designed to be played. It is unwieldy and somewhat mindbending to keep track of what is happening at any given point, particularly given the lack of screen. The visual feedback is extremely limited, forcing the player to focus on the sound. The existence of all of the controls can also lead to problematic, or at least unintended results. The pitch knob, for example, changes the playback speed of the current sound buffer. However if the “permanence" switch is activated, the overwriting of the buffer will still occur at the original playback rate (due to limitations in the audio library used for writing signals). It is virtually impossible to anticipate the outcome of this mismatch, as it depends on a complex relationship between the relative timings.

Nevertheless, with repeated use it becomes possible to reach a level of understanding to the point where a player can improvise a performance from an audio source. The instrument acts as something in between a looper and an effect, and the act of destruction can be as permanent or as temporary as desired. If the permanence switch has not been activated, the reset button can bring the original sound back from apparent death. Or, with some care, a destroyed sound can be “reset" to a semi-destroyed state and re-destroyed in a different fashion. The aesthetics of the destruction can be adjusted in real time; a high value “height" combined with a short “length" will generate noticeable but short glitches, while the opposite gives the potential for large sections of the sound to be momentarily slowed to a crawl before continuing at the original rate.

I say “a player", with an apparently impartial tone, but the reality is I am the only one to have played it. It remains to be seen whether this design, and the aesthetic choices embedded in the destruction engine, would be equally expressive and satisfying to someone else. I suspect that the instrument would function best with a player that has had the system explained to them, though perhaps my own knowledge is in fact holding back my ability to perform with it in an expansive way.

4 FINAL THOUGHTS

Sound is a time-inefficient medium. It is much faster for you to read this sentence than it would be for me to read it out loud. Certainly the same is true of this method for destroying a sound - it would be far quicker to simply delete the file with the press of a button rather than listen to it slowly decay. However this project has never been about efficiency, but is rather built on sustainability concepts of degrowth, anti-cornucopian design, social justice, and care. As such the destruction engine and the three manifestations of the code into objects suggest three different approaches to deletion and destruction that apply these concepts to different degrees whilst eschewing solutionist tendencies of optimization.

From a purely material standpoint, it is also worth acknowledging that these processes of “destruction" do not include any material change. Although the scales are unimaginably small, the same amount of physical space is being taken up by the audio buffers, whether or not the content within them has been distorted beyond all recognition. If a 3 second audio file is in the Memory Destroyers, an audio buffer of 132300 points is holding that sound, and each glitch simply shifts where those points are without shrinking the overall size. While this work has never claimed to provide a solution to the constantly growing cloud and the illusion of infinite resources, from a design perspective it is important to keep this in mind and consider what this represents. If we compare the destruction engine to a more analog process, the equivalent could be to pass a magnet over a section of audio tape. This makes some sense from a sustainability point of view, as the tape (or, indeed, the digital disk space) is thus made available for re-use, whereas any other method of destruction - perhaps cutting and destroying - would necessarily involve some form of waste (slash and burn, one could say). Where projects such at ChronoTape [4] and The Disintegration Loops [2] have taken advantage of the materiality of tape in particular, the Destruction Machines focus on how this can be approached with digital formats.

Indeed, what can actually be destroyed? This work separates out the need for material degrowth and destruction from the experiential act of deleting and letting go. On an emotional level it seems callous and insensitive to discuss destruction in such a casual way. When describing this work to others it is not unusual for people to react viscerally when faced the prospect of losing treasured media artifacts for the sake of some conceptual interaction design research project. A technology that kills a sound each time it is heard punishes the very act of listening and the instinct to preserve memories. However in doing so, this approach also positions a recorded sound as something that *can* be killed. It effectively breaks the concept of a recording as something that can be revisited at any point. Rather, a sound becomes in some way alive, and something that must be listened to with care.

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